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$$f(x, y, z) = \ln(1 + x^2 + y^2 - z^2) \text{ in } \mathbf{a}(1, -1, 1), \text{ richting } \mathbf{h}(2, -2, -3)$$

$$\frac{\partial f}{\partial x} = \frac{2x}{1 + x^2 + y^2 - z^2}$$

$$\frac{\partial f}{\partial y} = \frac{2y}{1 + x^2 + y^2 - z^2}$$

$$\frac{\partial f}{\partial z} = \frac{-2z}{1 + x^2 + y^2 - z^2}$$

Noemer in $\mathbf{a}$: $1 + 1 + 1 - 1 = 2$, dus:

$$Df(\mathbf{a}) = \left(\frac{2}{2} \quad \frac{-2}{2} \quad \frac{-2}{2} \right) = (1 \quad -1 \quad -1)$$

$$Df(\mathbf{a}, \mathbf{h}) = Df(\mathbf{a}) \cdot \mathbf{h} = (1 \quad -1 \quad -1) \cdot \begin{pmatrix} 2 \\ -2 \\ -3 \end{pmatrix} = 2 + 2 + 3 = 7$$

References